

Some useful mathematics

A

A.1 Vector and matrix notations

We frequently use vector and matrix notation in this book as it is extremely convenient for specifying neural network models. It is a shorthand notation for otherwise lengthy looking formulas, and formulas written in this notation can easily be entered into MATLAB. We consider three basic data types:

(1) Scalar:

$$a \text{ for example } 41 \quad (\text{A.1})$$

(2) Vector:

$$\mathbf{a} \text{ or component-wise } \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} \text{ for example } \begin{pmatrix} 41 \\ 7 \\ 13 \end{pmatrix} \quad (\text{A.2})$$

(3) Matrix:

$$\mathbf{a} \text{ or component-wise } \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \\ a_{31} & a_{32} \end{pmatrix} \text{ for example } \begin{pmatrix} 41 & 12 \\ 7 & 45 \\ 13 & 9 \end{pmatrix} \quad (\text{A.3})$$

We used bold face to indicate both a vector and a matrix because the difference is usually apparent from the circumstances. A matrix is just a collection of scalars or vectors. We talk about an $n \times m$ matrix where n is the number of rows and m is the number of columns. A scalar is thus a 1×1 matrix, and a vector of length n can be considered an $n \times 1$ matrix. A similar collection of data is called *array* in computer science. However, a matrix is different because we also define operations on these data collections. The rules of calculating with matrices can be applied to scalars and vectors.

We define how to add and multiply two matrices so that we can use them in algebraic equations. The *sum of two matrices* is defined as the sum of the individual components

$$(\mathbf{a} + \mathbf{b})_{ij} = a_{ij} + b_{ij}. \quad (\text{A.4})$$

For example, $\mathbf{a}$ and $\mathbf{b}$ are 3×2 matrices, then

$$\mathbf{a} + \mathbf{b} = \begin{pmatrix} a_{11} + b_{11} & a_{12} + b_{12} \\ a_{21} + b_{21} & a_{22} + b_{22} \\ a_{31} + b_{31} & a_{32} + b_{32} \end{pmatrix} \quad (\text{A.5})$$

Matrix multiplication is defined as

$$(\mathbf{a} * \mathbf{b})_{ij} = \sum_k a_{ik} b_{kj}. \quad (\text{A.6})$$

A.1 Vector and matrix notations	323
A.2 Distance measures	325
A.3 The δ -function	326

The matrix multiplication is hence only defined as multiplication matrices $\mathbf{a}$ and $\mathbf{b}$ where the number of columns of the matrix $\mathbf{a}$ is equal to the number of rows of matrix $\mathbf{b}$. For example, for two square matrices with two rows and two columns, their product is given by

$$\mathbf{a} * \mathbf{b} = \begin{pmatrix} a_{11}b_{11} + a_{12}b_{21} & a_{11}b_{12} + a_{12}b_{22} \\ a_{21}b_{11} + a_{22}b_{21} & a_{21}b_{12} + a_{22}b_{22} \end{pmatrix} \quad (\text{A.7})$$

A handy rule for matrix multiplications is illustrated in Fig. A.1. Each component in the resulting matrix is calculated from the sum of two multiplicative terms. The rule for multiplying two matrices is tedious but straightforward and can easily be implemented in a computer. It is the default when multiplying variables of the matrix type in MATLAB. If we want to multiply each component of a matrix by the corresponding component in a second matrix, we just have to include the operator `.*` between the matrices in MATLAB where the dot in front of the multiplication sign indicates 'component-wise'.

$$\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix} = \begin{pmatrix} a_{11}b_{11} + a_{12}b_{21} & a_{11}b_{12} + a_{12}b_{22} \\ a_{21}b_{11} + a_{22}b_{21} & a_{21}b_{12} + a_{22}b_{22} \end{pmatrix}$$

Fig. A.1 Illustration of a matrix multiplication. Each element in the resulting matrix consists of terms that are taken from the corresponding row of the first matrix and column of the second matrix. Thus in the example we calculate the highlighted element from the components of the first row of the first matrix and the second column of the second matrix. From these rows and columns we add all the terms that consist of the element-wise multiplication of the terms.

Another useful definition is the *transpose* of a matrix. This operation, indicated usually by a superscript t or a prime ($'$). The latter is used in MATLAB. Taking the transpose of a matrix means that the matrix is rotated 90 degrees; the first row becomes the first column, the second row becomes the second column, etc. For example, the transpose of the example in A.3 is

$$\mathbf{a}' = \begin{pmatrix} 41 & 7 & 13 \\ 12 & 45 & 9 \end{pmatrix} \quad (\text{A.8})$$

The transpose of a vector transforms a column vector into a row vector and vice versa.

As already mentioned, matrices were invented to simplify the notations for systems of coupled algebraic equations. Consider, for example, the system of three equations

$$41x_1 + 12x_2 = 17 \quad (\text{A.9})$$

$$7x_1 + 45x_2 = -83 \quad (\text{A.10})$$

$$13x_1 + 9x_2 = -5. \quad (\text{A.11})$$

This can be written as

$$\mathbf{a}\mathbf{x} = \mathbf{b} \quad (\text{A.12})$$

with the matrix $\mathbf{a}$ and vector $\mathbf{b} = (17, -83, -5)^T$ by using matrix operations in MATLAB.

A.2 Distances

We are often comparing two vectors. The distance d , on the difference between them, are many possible. We should improve the definition.

- The value of the distance is negative.
- The value of the distance is positive.
- The distance is zero.

We sometimes use the distance d , in which the distance is the number of steps, $\{0, 1\}$. A common distance, d^h , is the distance between two vectors, N , the distance can be calculated.

However, it is not a distance, it is a number of steps, among the coordinates, *a priori*. One can define a distance between two vectors. The distance is given by

We used the distance of a matrix to the geometrical interpretation of the distance to the cosine

where $\|\mathbf{a}\|$ is the norm of the vector $\mathbf{a}$, is defined as

with the matrix $\mathbf{a}$ as in the example of A.3, the vector $\mathbf{x} = (x_1 \ x_2)'$, and the vector $\mathbf{b} = (17 \ -83 \ -5)'$. Solutions of such equation systems can be formulated by using matrix notation, and many corresponding routines are implemented in MATLAB.

A.2 Distance measures

We are often confronted with comparing two vectors. It is straightforward comparing two vectors to decide if they are the same by inspecting the equivalence of each component. However, if they are different we would like to put a value, d , on the difference that gives us some measure of how different they are. There are many possible different definitions of such a difference measure, although we should impose some obvious restrictions on such measures:

- The value should be $d = 0$ if the vectors are the same. We do not allow negative values because we wish to interpret this value as a distance.
- The value should be positive, $d > 0$, if they are not the same.
- The distance from vector $\mathbf{a}$ to $\mathbf{b}$ should be the same as from $\mathbf{b}$ to $\mathbf{a}$.

We sometimes have to evaluate the distance between two binary vectors $\mathbf{a}$ and $\mathbf{b}$, in which the components have only two possible values for example $a_i, b_i \in \{0, 1\}$. A common value for the distance of such vectors is obtained by counting how many components (bits) are different. This measure is called the *Hamming distance*, d^h . If we normalize this value to the number of components of the vector, N , then we get a value between 0 and 1. This *normalized Hamming distance* can be calculated using the formula

$$d^h(\mathbf{a}, \mathbf{b}) = \frac{1}{N} \sum_i a_i(1 - b_i) + (1 - a_i)b_i. \quad (\text{A.13})$$

However, it is not obvious how to generalize this measure to vectors with real numbered components as we would have to judge the amount of the difference among the components, using their possible values, which we might not know *a priori*. One possible definition is to use the *normalized dot product* between vectors. The *dot product*, or *inner product*, of two column vectors $\mathbf{a}$ and $\mathbf{b}$ is given by

$$\mathbf{a}'\mathbf{b} = \sum_i a_i b_i. \quad (\text{A.14})$$

We used the transpose of the second vector so that this is only a special case of a matrix multiplication with a row vector $\mathbf{a}'$ and a column vector $\mathbf{b}$. In a geometrical interpretation of vectors (see Fig. A.2) this number is proportional to the cosine of the angle between the two vectors,

$$\mathbf{a}\mathbf{b}' = \|\mathbf{a}\| \|\mathbf{b}\| \cos(\alpha), \quad (\text{A.15})$$

where $\|\mathbf{a}\|$ is the *length* or *norm* of vector $\mathbf{a}$. The *Euclidean norm* of the vector is defined as

$$\|\mathbf{a}\| = \sqrt{\sum_i a_i^2}, \quad (\text{A.16})$$

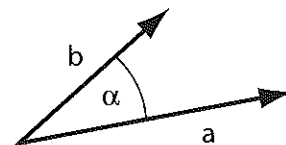


Fig. A.2 Graphical representation of two vectors in a two-dimensional space with an angle α between them.

which can also be written with the help of a dot product as

$$\|a\| = \sqrt{a'a}. \quad (\text{A.17})$$

The cosine of the angle α between two vectors is a number between -1 and 1 that is only zero when the vectors are pointing in the same direction. The absolute value of this number is therefore a possible definition of the similarity of two vectors,

$$d^{\text{dot}} = \left| \frac{ab'}{\|a\| \|b'\|} \right|. \quad (\text{A.18})$$

Other definitions, such as taking the square of the normalized dot product, are also valid. Another definition that is sometimes used as measure of the distance between two vectors with positive real numbered components $a_i, b_i \in \mathbf{R}^+$ has the form of the *Pearson correlation coefficient* from statistics, namely

$$d^{\text{Pearson}} = \frac{\|ab\| - \|a\| \|b\|}{\sqrt{(\|a^2\| - \|a\|^2)(\|b^2\| - \|b\|^2)}}. \quad (\text{A.19})$$

We call this distance measure the *Pearson distance*.

A.3 The δ -function

The δ -function is a very convenient notation, which is formally a functional since it is only defined as an operation over a function. One can think of it as a density function that is zero except for its arguments for which it is infinite, and the integral over the δ -function is one; that is,

$$\int_{-\infty}^{\infty} \delta(x - x_1) dx = 1. \quad (\text{A.20})$$

It is usually used as an integration kernel with other functions as in

$$\int_{-\infty}^{\infty} \delta(x_1) f(x) dx = f(x_1). \quad (\text{A.21})$$

The delta function is useful for writing discrete events in a continuous form.